

*MA.8.AR.2.1***Benchmark**

MA.8.AR.2.1 Solve multi-step linear equations in one variable, with rational number coefficients. Include equations with variables on both sides.

Benchmark Clarifications:

Clarification 1: Problem types include examples of one-variable linear equations that generate one solution, infinitely many solutions or no solution..

Connecting Benchmarks/Horizontal Alignment

- MA.8.NSO.1.7
- MA.8.AR.1.3
- MA.8.AR.4.1
- MA.8.GR.1.4, MA.8.GR.1.5, MA.8.GR.1.6

Terms from the K-12 Glossary

- Coefficient
- Linear Equation
- Rational Number

Vertical Alignment**Previous Benchmarks**

- MA.7.AR.2.2

Next Benchmarks

- MA.912.AR.2.1

Purpose and Instructional Strategies

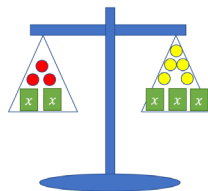
In grade 7, students wrote and solved two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers. In grade 8, students solve multi-step linear equations in one variable, with rational number coefficients, including equations with variables on both sides. In Algebra 1, students will write and solve linear equations in one variable in a real-world context, with rational number coefficients.

- In this benchmark, students work with linear equations, which is foundational for the work with both linear equations and nonlinear equations throughout all future mathematics courses.

- Instruction includes the use of manipulatives, drawings, models, properties of operations and properties of equality.
 - Algebra Tiles

$$3x - 12 = -2x + 13$$

- Balance



- Problem types involve multi-step problems that require the use of the distributive property, combining like terms, and variables on both sides of the equation.
- Since there are variables on both sides of the equation, instruction includes discovering that one-variable equations can result in three possible solution sets. The possible solutions are one solution, no solution or infinitely many solutions. This benchmark provides a foundation for MA.8.AR.4 when students are working with systems of equations and two-variable equations.

Common Misconceptions or Errors

- Students may incorrectly apply the distributive property by multiplying the monomial to only one of the terms in the parentheses. To address this misconception, emphasize that it is the distributive property of multiplication over addition to help support student understanding.
- Students may incorrectly apply the rules of integer arithmetic as they distribute when working with the operations of negative numbers and applying the distributive property of multiplication over addition.
- Students may incorrectly think that you always need a constant as a solution when given a variable. To address this misconception, provide examples that show a constant equal to a variable as a solution, a constant equal to a constant or a false equality statement.

Strategies to Support Tiered Instruction

- Teacher provides opportunities to demonstrate the distributive property as repeated addition of the given expression.
 - For example, when using algebra tiles, the expression $3(x - 4)$ can be represented as adding $(x - 4)$ three times together.



Or

$$3(x - 4) = (x - 4) + (x - 4) + (x - 4) = 3x - 12$$

- Teacher co-creates a graphic organizer with examples of one, no solutions, and infinitely many solutions. Demonstrate using substitution to help students make sense of the solutions.
- Teacher co-creates an anchor chart for multiplying negative integers for students that incorrectly apply the rules of negative integers as they distribute.
- Teacher provides examples for students that need additional support for distributive property by using the area model for $2(x + 4)$.

	x	4
2	$2x$	8

- Instruction includes emphasizing that it is the distributive property of multiplication over addition to help support student understanding.

Instructional Tasks

Instructional Task 1 (MTR.1.1, MTR.4.1)

Part A. How many solutions does the equation, $2y + 7 = 7 + 2y$ have? Explain your reasoning to another student and justify your answer.

Part B. How many solutions does the equation, $2(y + 3) + 1 = 2(3.5 + y)$ have? Explain your reasoning to another student and justify your answer.

Part C. What do you notice about the equations in Part A and Part B?

Instructional Task 2 (MTR.1.1, MTR.4.1)

For each equation, state whether there is no solution, one solution, or infinitely many solutions. Explain your reasoning.

Equation	Number of Solutions	Explanation
$\frac{1}{3}x - 6 = \frac{1}{3}(x - 3) - 1$		
$2x + 7 = -2x + 7$		
$0.8x + 1.4 = 0.8x$		

Instructional Items

Instructional Item 1

Solve for x :

- $-3.5(10x - 2) = -176.75$
- $15(2x - 10) + 4x = -3(15x + 4)$

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*